

0900–1200 hrs

6 / Nov / 2023

DJ Exam Hall

EXAMS OFFICE
USE ONLY

University of the Witwatersrand, Johannesburg

Course or topic No(s)

COMS4033A/7044A

Course or topic name(s)
Paper Number & title

Artificial Intelligence

Examination to be held
during the month(s) of

November, 2023

Year of study

Degrees/Diplomas for
which this course is
prescribed

Faculties presenting can-
didates

Science

Internal examiner(s)

Dr Steven James
x-76157

External examiner(s)

Prof. George Konidaris (Brown)

Special materials

None

Time allowance

3 Hours

Instructions to candidates

70 Marks available. 70 marks = 100%.
Answer all questions. This is a closed book exam.
This exam consists of 14 pages.

Artificial Intelligence

COMS4033A/7044A

Student Number: _____ Row: _____ Seat: _____

For marking purposes only

Question 1	
Question 2	
Question 3	
Question 4	
Question 5	
Total	

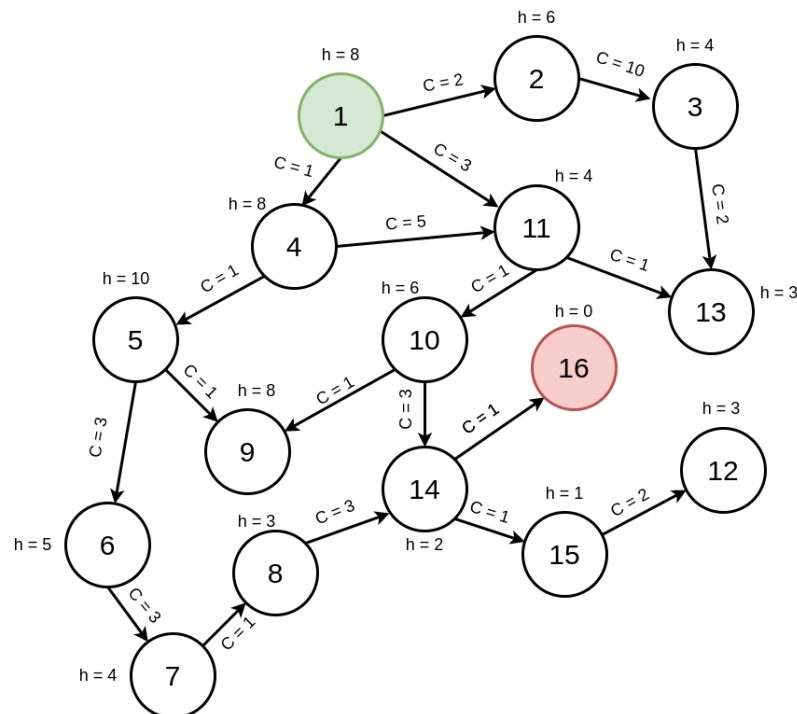
Instructions

- Answer all questions in pen. **Do not write in pencil.**
- This test consists of 14 pages. Ensure that you are not missing any pages.
- This is a **closed-book** test: you may not consult any written material or notes.
- You are allocated 3 hours to complete this exam.
- There are 5 questions and 70 marks available.
- Ensure your cellphone is switched off.

Question 1**Agents and Search****[18 Marks]**

1. What is the difference between the strong AI hypothesis and the weak AI hypothesis? [1]

2. Let C be the cost of traversing an edge between nodes, and let h be the heuristic estimate of a given node to the goal. Consider the graph below, where nodes are labelled with integers and directed edges are indicated by lines with arrows. Each edge is labelled with a cost C , and each node with a heuristic value h .



Assume that we wish to search for the goal node 16 starting at node 1 using the greedy best first search (GBFS) algorithm. Further, assume that if there are ties in the priority queue, they are resolved by removing the node with the smallest label first (e.g. if two nodes 2 and 1 both have priority 10, then 1 will be removed before 2).

Given this, detail each step of the GBFS algorithm. At each iteration, describe what node is popped from the queue and then what nodes are added to the queue subsequently (and their corresponding priority). [4]

3. Consider the game of reconnaissance blind chess as described in the assignment. We can categorise this task as follows:

✓

- (a) partially observable or fully observable
- (b) deterministic or stochastic
- (c) episodic or sequential
- (d) static or dynamic
- (e) discrete or continuous
- (f) single agent or multiagent

For each option above, circle the correct one.

[3]

✓

4. List one advantage of depth-first search over breadth-first search?

[1]

5. Consider the following payoff matrix representing a game between two players A (the “row” player) and B (the “column” player).

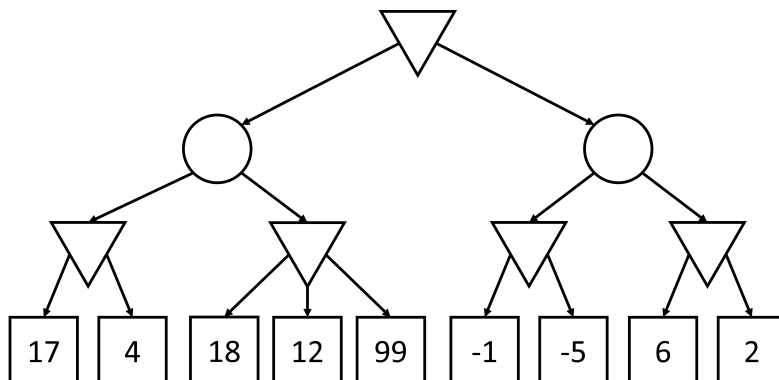
	B_1	B_2	B_3	B_4
A_1	(1, 7)	(2, 5)	(7, 2)	(0, 1)
A_2	(5, 2)	(3, 3)	(5, 2)	(3, 1)
A_3	(7, 0)	(2, 5)	(0, 4)	(1, 1)
A_4	(0, 0)	(0, 2)	(0, 0)	(0, -1)

- (a) Execute the *maximin* algorithm on the payoff matrix to find the outcome, beginning with player *A*. At each step, list the strategy selected by the relevant player, as well as the final outcome. [2]

- (b) Repeat the above, but now begin with player *B*. What is the outcome of this procedure? [1]

- (c) What can be said about the strategies produced by the *maximin* algorithm for both players? Motivate your answer. [1]

6. Consider the game tree below. Triangles represent the maximising player, circles the minimising player and rectangles terminal nodes. Execute alpha-beta pruning on the tree. Annotate the tree by writing down the value backed up to each node in the tree (including the root), and indicate where pruning takes place by crossing out the relevant edges. [5]



Question 2 Knowledge Representation [9 Marks]

1. Consider a knowledge base with three objects A, B and C. Instantiate the following sentence: $\forall x, Human(x) \Rightarrow Mortal(x)$. [1]

2. Alice wishes to encode the following sentence in first-order logic: "John does not love anyone" She writes this as $\exists x \neg \text{love}(\text{John}, x)$. Her friend Bob disagrees with her and thinks it should be written as $\neg \forall x \text{love}(\text{John}, x)$. Eve disagrees with them both and thinks it should be $\neg \exists x \text{love}(\text{John}, x)$. Explain who is correct and provide a reason why the other two are incorrect. [3]

3. Assume the probability of having some disease is 0.0005, and a test for this disease is 99% accurate. What is the probability one has this disease if one tests positive for it? [2]

4. A patient has been suffering from shortness of breath (called dyspnoea) and visits the doctor, worried that he has lung cancer and he has read that one of the symptoms of lung cancer is dyspnoea. The doctor knows that the chance of the patient having cancer (or some other disease) is affected by whether he is a smoker or has been exposed to air pollution. A positive X-ray would indicate lung cancer. We wish to model this using a Bayesian network with the following random variables: whether the patient has been exposed to pollution (*pollution*), whether they are a smoker (*smoker*), whether they have cancer (*cancer*) or dyspnoea (*dyspnoea*) and whether the X-ray returns a positive cancer diagnosis (*xray*).

(a) Draw the corresponding Bayesian network for the above three variables. [2]

(b) Write down the factorisation of joint probability distribution given the Bayesian network's structure in part a). [1]

Question 3**Planning****[23 Marks]**

- ✓ 1. What is the fundamental difference between search and STRIPS planning? [1]

- ✓ 2. What is the difference between a PDDL precondition and effect? [2]

- ✓ 3. List the components that make up a PDDL problem description. [3]

- ✓ 4. Describe how the FF planner creates a relaxed version of a planning problem [2]

- ✓ 5. Consider the Blocksworld environment with ten blocks labelled A–J. The current state is given by

```
(clear C) (clear F) (ontable I) (ontable F) (on C E) (on E J)
(on J B) (on B G) (on G H) (on H A) (on A D) (on D I) (handempty))
```

and there is an operator `unstack`, which is defined as follows:

```
(:action unstack
  :parameters (?x ?y)
  :precondition (and (on ?x ?y) (clear ?x) (handempty))
  :effect
  (and (holding ?x)
    (clear ?y)
    (not (clear ?x))
    (not (handempty))
    (not (on ?x ?y))))
```

Write down the new state that results from the execution of `unstack(c, e)`. [3]

- ✓ 6. Under what conditions do we expect that regression planning would outperform forward planning? [1]

- ↓ ✗ 7. Define what is meant by a *macro-action*. [1]

- ✓ 8. Name two things that make a scheduling problem harder than a planning problem. [2]

- ✓ 9. What are the five components that make up a Markov decision process? Explain what each of them are. [3]

- ↓ ✗ 10. In words, define what is meant by the *return*. [1]

✓ 11. What is the difference between a policy and a plan? [1]

✓ ~~12.~~ 12. When estimating a value function for a policy, why is Monte Carlo considered to be high variance? [1]

✓ 13. How does prioritised sweeping improve on policy iteration? [2]

Question 4**Reinforcement Learning****[8 Marks]**

1. Explain how Q-learning overcomes the challenge of having to act greedily with respect to a value function. [1]



2. Describe what is meant by the exploration-exploitation dilemma. [2]



3. Write down the SARSA update rule. How does this differ from the Q-learning update rule? [3]



4. What is the main difference between early (pre 2000) attempts at function approximation, and function approximation using deep learning (with neural networks?) [1]



5. Describe how the DQN algorithm overcomes the problem training using data that is highly correlated. [1]



Question 5**Machine Learning****[12 Marks]**

1. With regards to regression, describe how overfitting occurs and why it is problematic. [2]

2. Describe the basic idea behind k -fold cross validation. [1]

3. How does a decision tree learning algorithm decide which attribute to test next? [2]

4. Describe how the *bandwidth* parameter in kernel density estimation relates to overfitting and underfitting. [2]

5. Describe how autoencoders can be pretrained for downstream classification tasks. [2]

- ✓ 6. Consider the following datapoints: $x_1 = (2, 10)$, $x_2 = (2, 5)$, $y_1 = (5, 8)$, $y_2 = (7, 5)$, $z_1 = (1, 2)$, $z_2 = (4, 9)$. We wish to perform k -means clustering with a Euclidean distance metric on this data with $k = 3$. The initial centroids are chosen to be x_1, y_1, z_1 . Given this, write down all steps of the algorithm until convergence. Your answer should include distance calculations, as well as any updates to the centroids and the labelling of datapoints to their clusters. [3]

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